

Mugahed Izzeldin

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SUMMARY

PhD Candidate in Computer Science and systems engineer with experience building reproducible infrastructure, distributed systems, cloud-native platforms, and production mobile network systems. Strong background in Linux, C, Go, Python, Kubernetes, Nix/NixOS, networking, and operational troubleshooting. Owner and maintainer of open-source systems projects including NixCore, with research spanning mobile core networks, cloud computing, and reproducible systems. Experienced taking systems from design and implementation through evaluation, documentation, operations, and cross-layer debugging.

TECHNICAL SKILLS

Networking: TCP/IP, BGP, OSPF, MPLS, QoS, IPsec, VPNs, L2/L3 switching and routing, 2G/3G/4G/5G packet core

Programming: C, Go, Python, JavaScript

Systems: Linux, Nix/NixOS, distributed systems, Raft consensus, key-value stores, reproducible infrastructure

Cloud & Infrastructure: Docker, Kubernetes, Helm, Ansible, multi-tenant cloud operations

Databases: PostgreSQL, MySQL, MongoDB

Tools & Operations: Git, CI/CD concepts, monitoring, production troubleshooting, technical documentation

WORK EXPERIENCE

Graduate Research Assistant | School of Computing – University of Utah | Aug 2021 – Present

- Conduct systems research in the Flux Research Group under Prof. Robert Ricci, focused on cloud computing, mobile networks, reproducible infrastructure, and networked systems.
- Build and evaluate software infrastructure for mobile core research, including reproducible deployment workflows using Linux, Nix/NixOS, Kubernetes, and containerized network components.
- Own research software across design, implementation, experiment execution, debugging, documentation, and publication.
- Support CloudLab-related research and infrastructure activities, working in a large-scale experimental cloud environment used by distributed systems and networking researchers.
- Communicate research through technical writing, papers, experiments, and collaboration with academic and industry partners.

5G Core Network Research Engineer | Celona Inc. | May 2022 – Aug 2022

- Supported research and engineering work for Celona's private 5G network solutions.
- Analyzed commercial 5G core behavior in private-network deployments, contributing to performance evaluation and publication-quality research.
- Worked across mobile core, networking, and cloud deployment layers to investigate system behavior and deployment tradeoffs.

Core Network Engineer | Atlas Telecommunication | Mar 2021 – Aug 2021

- Managed LTE core, IP networking, and transmission operations for a private LTE network, with end-to-end responsibility across core, transport, and access-network interfaces.
- Troubleshot production issues across packet core, routing, and transmission layers, coordinating fixes across technical teams.

Core & MPBN Back-Office Engineer | Huawei Technologies Sudan | Oct 2017 – Mar 2021

- Operated and supported mobile packet core and IP backbone systems for service-provider networks, including core nodes, routing infrastructure, firewalls, and customer-facing services.
- Configured, tested, launched, and troubleshot GPRS roaming and integrations between core network elements and 2G/3G/4G radio access networks.
- Generated and analyzed packet-core and IP backbone KPI reports to identify, troubleshoot, and resolve service-impacting issues.
- Implemented new network solutions, handled production emergencies, and expanded customer resources to improve capacity and service availability.
- Designed and developed internal tools for packet-core documentation, reporting, and analysis.

Core Network Planning and Optimization Engineer | MTN Sudan | Feb 2017 – Oct 2017

- Designed, dimensioned, and documented packet-core and IP-core network changes for a mobile service-provider environment.
- Monitored service KPIs and coordinated optimization work to maintain network quality and reliability.
- Integrated new marketing services and data bundles into packet-core systems while documenting change requests and operational procedures.

SELECTED SYSTEMS & OPEN-SOURCE PROJECTS

NixCore Framework | Project Owner | nixcore.org

- Designed and maintain NixCore, a reproducible mobile network core framework for deploying, configuring, and evaluating 4G/5G core network components using Nix/NixOS, Kubernetes, and declarative infrastructure.
- Built the framework to support repeatable research environments, reduce deployment drift, and make mobile-core experimentation easier to reproduce and debug.
- Own project direction, implementation, documentation, and ongoing maintenance as an open-source research framework.
- Developed NixCore as the foundation for an ongoing research paper, “NixCore: Toward a Reproducible Mobile Network Core.”

pomo Nix Package | Maintainer | [NixOS package](#)

- Maintain the pomo package in Nixpkgs, contributing to open-source packaging, reproducible builds, version updates, and package quality for NixOS users.

Missing Semester Arabic Website | Maintainer | missing-semester-ar.github.io

- Maintain the Arabic version of MIT's Missing Semester course website, supporting accessible technical education for Arabic-speaking students and engineers.

Raft-Based Key-Value Store

- Implemented a distributed key-value store based on the Raft consensus algorithm, covering replicated state, leader-based coordination, and fault-tolerant system behavior.

Production Network Infrastructure Projects

- Designed and implemented production network services including DNS, RADIUS authentication, internet filtering, IPv6 pilot design, and GRX design for ISP/mobile-core environments.
- Participated in MTN Sudan's IP core swap and CloudEdge integration, including hardware expansion and VNF scale-out for packet-core infrastructure.

EDUCATION

PhD in Computer Science | University of Utah | Expected Jun 2027

Research areas: systems, cloud computing, reproducible infrastructure, and mobile networks.

SELECTED PUBLICATIONS

- “RaaS: How to Fix Mobile Roaming” Mugahed Izzeldin, Ryan W. West, Mary Ogunmola, and Robert Ricci. Manuscript under review, 2026.
- “NixCore: Toward a Reproducible Mobile Network Core” Mugahed Izzeldin and Robert Ricci. Manuscript under review, 2026.
- “Analysing the performance of commercial 5GC in private network deployment” Mugahed Izzeldin, Lyutianyang Zhang, Vanlin Sathya, Mohit Goyal, and Mehmet Yavuz. International Journal of Mobile Network Design and Innovation, 2024.
- “Approaches for SMS encryption and user accounts verification” M. I. O. Hajahmed, K. E. M. Osman, and O. T. M. Ali. ICCCEEE20 Conference, 2021.
- “Confidential5G Trusted Execution of the 5G Core on Untrusted Clouds” Mary Ogunmola, Mugahed Izzeldin, Robert Ricci. Manuscript under review, 2026.
- “Implementation of SDN in a Campus NAC Use Case” M. I. O. Hajahmed, M. Adil, M. Omar, M. Khalid.. B.Sc. (Honors) degree project.

PROFESSIONAL DEVELOPMENT & SERVICE

- Attended ACM CCS 2024 and KubeCon + CloudNativeCon North America 2024.
- Certifications and coursework: Huawei HCIA Routing and Switching, CCNP ROUTE, CCNA, IPv6 workshop by AFRINIC, SQL, Python, data analysis, and web development.
- Volunteer service: IEEE Sudan Subsection Media Coordinator, IEEEExtreme proctor, IEEE SUST branch election organizer, and technical seminar presenter.